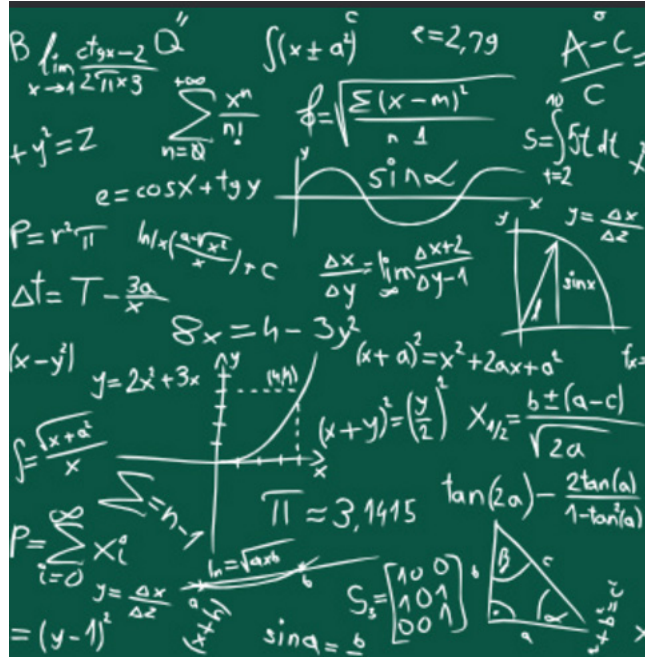


Last update: September 26, 2026

Exercises: verifying limits of sequences



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Using the definition of limit, verify that:

$$\lim_{n \rightarrow +\infty} \frac{n}{2n+5} = \frac{1}{2}$$

We have

$$a_n = \frac{n}{2n+5}$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow \frac{1}{2} - \varepsilon < a_n < \frac{1}{2} + \varepsilon$$

We have

$$a_n - \frac{1}{2} = -\frac{5}{4n+10},$$

hence we must verify that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < -\frac{5}{4n+10} < \varepsilon$$

The second inequality is always true, while the first is equivalent to

$$4n+10 > \frac{5}{\varepsilon} \text{ hence it is satisfied for } n > \frac{5}{4\varepsilon} - \frac{5}{2}$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{5}{4\varepsilon} - \frac{5}{2}$$

to satisfy the condition required by the definition of limit.

Exercise 2

Using the definition of limit, verify that:

$$\lim_{n \rightarrow +\infty} \frac{1}{\sqrt{n+1}} = 0$$

We have

$$a_n = \frac{1}{\sqrt{n+1}}$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < a_n < \varepsilon$$

The first inequality is always true, while the second is equivalent to

$$\sqrt{n+1} > \frac{1}{\varepsilon} \text{ hence it is satisfied for } n > \frac{1}{\varepsilon^2} - 1$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^2} - 1$$

to satisfy the condition required by the definition of limit.

Exercise 3

Using the definition of limit, verify that:

$$\lim_{n \rightarrow +\infty} \log\left(1 + \frac{1}{n}\right) = 0$$

We have

$$a_n = \log\left(1 + \frac{1}{n}\right)$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < a_n < \varepsilon$$

The first inequality is always true, while the second is equivalent to

$$1 + \frac{1}{n} < e^\varepsilon \text{ hence it is satisfied for } n > \frac{1}{e^\varepsilon - 1}$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{e^\varepsilon - 1}$$

to satisfy the condition required by the definition of limit.

Exercise 4

Using the definition of limit, verify that:

$$\lim_{n \rightarrow +\infty} \sqrt{4 + \frac{1}{n}} = 2$$

We have

$$a_n = \sqrt{4 + \frac{1}{n}}$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow 2 - \varepsilon < a_n < 2 + \varepsilon$$

The first inequality is always true since $a_n > 2$ for every n , while the second is equivalent to

$$4 + \frac{1}{n} < 4 + \varepsilon^2 + 4\varepsilon \quad \text{hence it is satisfied for } n > \frac{1}{\varepsilon^2 + 4\varepsilon}$$

For a fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^2 + 4\varepsilon}$$

to satisfy the condition required by the definition of limit.

Exercise 5

Using the definition of limit, verify that:

$$\lim_{n \rightarrow +\infty} (n^2 - 1) = +\infty$$

We have

$$a_n = n^2 - 1$$

We must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

$$n^2 - 1 > M \text{ is satisfied for } n > \sqrt{M+1}$$

Hence, for a fixed $M > 0$, it suffices to choose the first integer

$$n(M) > \sqrt{M+1}$$

to satisfy the required divergence condition.

Exercise 6

Using the definition of limit, verify that:

$$\lim_{n \rightarrow +\infty} \log(\sqrt{n} + 1) = +\infty$$

We have

$$a_n = \log(\sqrt{n} + 1)$$

We must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

$$\log(\sqrt{n} + 1) > M \text{ is satisfied for } n > (e^M - 1)^2$$

Hence, for a fixed $M > 0$, it suffices to choose the first integer

$$n(M) > (e^M - 1)^2$$

to satisfy the required divergence condition.